

Normal Structure & Fixed Point Theory

Chapter 6c: Applications of Monotone Operator Theory

Based on Pathak's *An Introduction to Nonlinear Analysis and Fixed Point Theory*

Lecture Series on Monte Carlo Theory

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Outline I

- 1 Introduction: Normal Structure and Fixed Points
- 2 Monotone Operators and Hilbert Spaces
- 3 Hammerstein Integral Equations
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What is Normal Structure?

Definition 1 (Normal Structure)

A convex set K in a Banach space X is said to have **normal structure** if for each bounded convex subset H of K that contains more than one point, there exists a point $x \in H$ that is not a diametral point of H .

- A diametral point is one where the distance to the boundary equals the diameter $\delta(H)$
- Normal structure is a fundamental concept in fixed point theory
- Every compact convex set in a Banach space has normal structure (Theorem 5.27)

Example 2

Any bounded convex set in a uniformly convex Banach space has normal structure.

Central Question:

When do nonexpansive and monotone mappings have fixed points?

Key Results:

- 1 Normal structure implies existence of fixed points for certain nonexpansive mappings
- 2 Kirk's theorem connects normal structure to fixed point properties
- 3 Monotone operators provide powerful tools for integral equations
- 4 Hammerstein equations: combining linear and nonlinear operators

This chapter bridges geometric properties (normal structure) with operator-theoretic methods.

Monotone Operators: Definition

Definition 3 (Monotone Operator)

An operator $T : X \rightarrow X^*$ on a Banach space X is **monotone** if for all $x, y \in X$:

$$\langle Tx - Ty, x - y \rangle \geq 0$$

Definition 4 (Strongly Monotone Operator)

T is **strongly monotone** with constant $c > 0$ if:

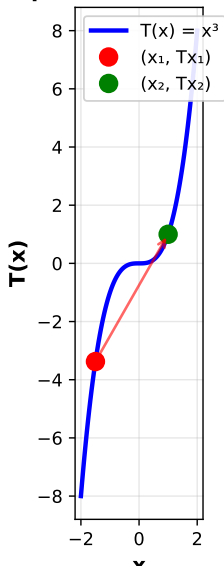
$$\langle Tx - Ty, x - y \rangle \geq c\|x - y\|^2$$

Examples:

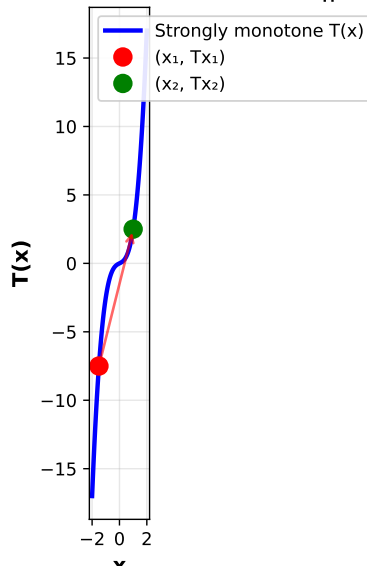
- Identity map I is strongly monotone with $c = 1$
- Gradient of convex function: ∇f is monotone if f is convex
- Integral operators with appropriate kernels

Monotone Properties

Monotone Operator: $(x_1 - x_2, T(x_1) - T(x_2)) \geq 0$



Strongly Monotone: $(x_1 - x_2, T(x_1) - T(x_2)) \geq c \|x_1 - x_2\|^2$



Key Theorem: Monotone Operators with Hilbert Spaces

Theorem 5 (Dolph-Minty, Theorem 8.8)

Let $\mathcal{H}$ be a Hilbert space and $N_f : \mathcal{H} \rightarrow \mathcal{H}$ be hemicontinuous and monotone, and $K : \mathcal{H} \rightarrow \mathcal{H}$ continuous and strongly monotone. Then:

- 1 Equation (8.31): $x + KN_fx = y$ has a **unique solution** x for every y
- 2 The solution depends continuously on y and satisfies:

$$\|x\| \leq \frac{1}{c} \|K\| \|y\|$$

where c is the strong monotonicity constant.

Proof idea: Use strong monotonicity to establish injectivity and surjectivity of the operator.

Applications: When K is Noncompact

Theorem 6 (Theorem 8.9: Monotone K with Coercivity)

Let $K : \mathcal{H} \rightarrow \mathcal{H}$ be linear, continuous, monotone, and $N_f : \mathcal{H} \rightarrow \mathcal{H}$ continuous, bounded, and monotone. If there exists $M > 0$ such that:

$$\langle N_f x, x \rangle \geq 0 \quad \text{for } \|x\| > M$$

Then equation (8.32): $x + KN_f x = 0$ has a solution.

Key insight: The coercivity condition prevents solutions from escaping to infinity.

Remark: Both theorems hold if the underlying space is a reflexive Banach space (not just Hilbert).

Hammerstein Integral Equation

Definition 7 (Nonlinear Hammerstein Integral Equation)

The problem of determining x satisfying:

$$x(s) + \int_{\Omega} k(s, t)f(t, x(t))dt = y(s), \quad s \in \Omega$$

where:

- $k : \Omega \times \Omega \rightarrow \mathbb{R}$ is a kernel function
- $f : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ is Carathéodory
- Ω is a measurable domain

This can be written as the operator equation:

$$x + KN_fx = y \quad (\text{Hammerstein form})$$

where:

- $K : [Kx](s) = \int_{\Omega} k(s, t)x(t)dt$ (linear operator)
- $N_f : [N_fx](s) = f(s, x(s))$ (nonlinear operator)

Hammerstein Equation Schematic

Hammerstein Integral Equation: $x + KNx = y$



- K: Bounded linear monotone operator (integral operator)
- N: Continuous, bounded, possibly monotone
- Solution x satisfies: $x + KNx = y$
- Solvability depends on monotonicity structure

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$

Example 8.6: Concrete Hammerstein Equation

Specific Hammerstein Problem

Consider the equation:

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = 0$$

with conditions:

- 1 $k(s, t)$ is symmetric, positive semidefinite, Hilbert-Schmidt kernel
- 2 $f(s, x)$ is monotone increasing in x with growth condition: $|f(s, x)| \leq a(s) + b|x|$
- 3 For large $|x|$: $f(s, x) > a + bx$ (for $x > M$) or $f(s, x) < -a + bx$ (for $x < -M$)

Result: This is equivalent to $x + KN_f x = 0$, and has a solution in $L_2(\Omega)$ by Theorem 8.9, because K is monotone and N_f satisfies the coercivity condition.

Numerical Example: Solving Hammerstein Equation

Python Implementation

```
import numpy as np
from scipy.integrate import quad
from scipy.optimize import fsolve

# Hammerstein equation parameters
def hammerstein_kernel(s, t):
    """Symmetric kernel:  $k(s, t) = \exp(-|s - t|)$ """
    return np.exp(-np.abs(s - t))

def nonlinear_term(t, x_t):
    """Nonlinear operator:  $f(t, x) = \tanh(x)$ """
    return np.tanh(x_t)

def hammerstein_operator(x_vals, s_eval):
    """Compute integral term of Hammerstein equation"""
    result, _ = quad(lambda t: hammerstein_kernel(s_eval, t) *
                    nonlinear_term(t, np.interp(t, s_pts, x_vals)),
                    0, 1)

    return result
```

Generalized Hammerstein Equations

Definition 8 (Generalized Hammerstein Equation)

The integral equation:

$$u(s) + \int_{\Omega} k(s, t, u(t)) f(t, u(t)) dt = v(s)$$

where the kernel $k(s, t, u)$ depends on the solution u .

Operator form: $u + K(u)N_f u = v$

Theorem 9 (Theorem 8.17: Generalized Equations)

Let X be a reflexive Banach space and for each $u \in X$:

- $K(u) : X^* \rightarrow X$ is angle-bounded with constant $c \geq 0$
- $u \mapsto K(u)$ is continuous from X to $L(X^*, X)$
- $\|K(u)\| \leq r$ for all $u \in X$

If $N_f : X \rightarrow X^*$ is continuous, bounded, and $(u_1 - u_2, N_f u_1 - N_f u_2) \geq -k\|u_1 - u_2\|^2$ with $kr < (1 + c^2)$, then $u + K(u)N_f u = 0$ has at least one solution.

Equations with Sum of Hammerstein Operators

Problem 10 (Mixed Type Integral Equation)

$$u(s) + \int_{\Omega} K_1(s, t) f_1(t, u(t)) dt + \int_{\Omega} K_2(s, t) f_2(t, u(t)) dt = 0$$

Operator form: $u + K_1 N_1 u + K_2 N_2 u = 0$

Key approach:

- 1 Define the product Banach space: $\mathcal{X} = X \times X \times \cdots \times X$
- 2 Define composite linear operator $K : \mathcal{X} \rightarrow \mathcal{X}^*$
- 3 Define composite nonlinear operator $N : \mathcal{X} \rightarrow \mathcal{X}^*$
- 4 Apply theorems to the combined equation

This allows handling systems of coupled integral equations through a unified framework.

Urysohn's Integral Equation

Definition 11 (Urysohn's Integral Equation)

An integral equation where f depends on both t and $x(t)$:

$$x(s) + \int_{\Omega} k(s, t, x(t)) dt = y(s)$$

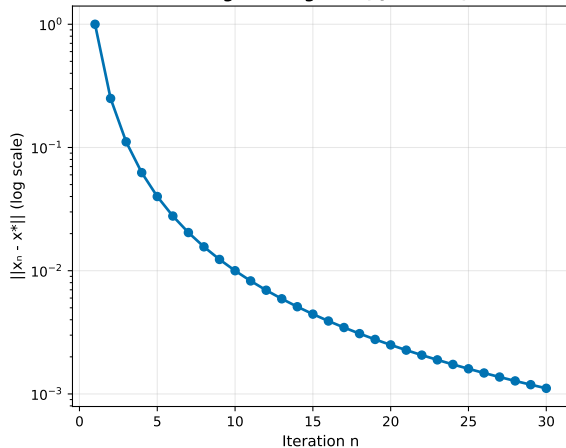
Characteristics:

- More general than Hammerstein (kernel depends on solution)
- Classical problem in nonlinear analysis
- Can be treated using monotone operator methods
- Often requires Leray-Schauder principle for existence

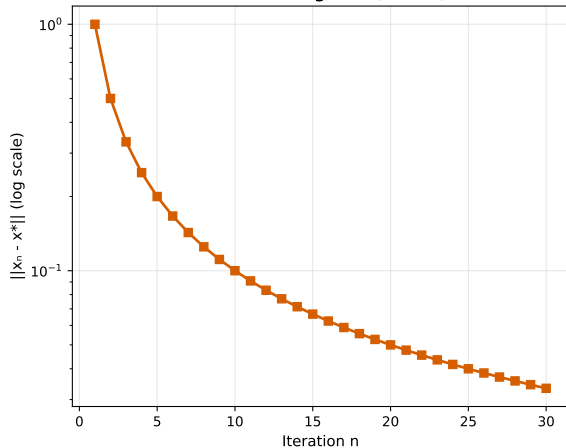
Remark: The monotone operator approach unifies treatment of different integral equation types through operator-theoretic perspective rather than specialized techniques for each type.

Iterative Methods for Hammerstein Equations

Strong Convergence (Quadratic)



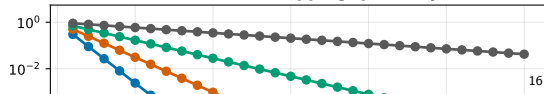
Weak Convergence (Linear)



Hammerstein Operator Convergence



Contraction Mapping: ρ^n Decay



Bruck's Jr. Iterative Scheme I

Theorem 12 (Theorem 8.21: Iterative Solution)

Let X be a reflexive Banach space and $K : X \rightarrow X^*$ a symmetric, monotone, bounded linear operator. Let $N : X^* \rightarrow X$ be a nonlinear hemicontinuous operator satisfying:

$$\langle u_1 - u_2, Nu_1 - Nu_2 \rangle \geq -k \|u_1 - u_2\|_{X^*}^2$$

If $k\|K\| < 1$, then $u + KNu = w$ has a unique solution $u \in X$, and the sequence defined by Bruck's scheme:

$$v_{n+1} = \frac{n}{n+1}v_n - \frac{1}{n+1}SN[S^*v_n + w]$$

converges to the solution with convergence rate $\|S^*v_n + w - u\| = O(n^{-1/2})$.

Bruck's Jr. Iterative Scheme II

Key Components:

- $S : X \rightarrow \mathcal{H}$ is a continuous linear mapping
- $\mathcal{H}$ is a Hilbert space with S^* injective
- The splitting lemma provides the decomposition
- Averaging mechanism $\frac{n}{n+1}v_n$ ensures convergence

Practical advantages:

- 1 Guaranteed convergence under verifiable conditions
- 2 Convergence rate $O(n^{-1/2})$ is optimal for this class
- 3 Applicable to wide range of nonlinear integral equations

Computational Scheme: Hammerstein Operator Equations

Algorithm: Iterative solver for $u + KNu = w$

- ① Initialize $v_1 \in \mathcal{H}$ (arbitrary)
- ② For $n = 1, 2, \dots, N_{\max}$:
 - ① Compute: $h_n = S^* v_n + w$
 - ② Evaluate: $N(h_n)$ (nonlinear operator)
 - ③ Apply: $K(u_n) = K(N^{-1} v_n) P_n N u_n$
 - ④ Update: $v_{n+1} = \frac{n}{n+1} v_n - \frac{1}{n+1} S N[S^* v_n + w]$
 - ⑤ If $\|v_{n+1} - v_n\| < \epsilon$, then break (convergence achieved)
- ③ Recover: $u = S^* v_n + w$

Convergence: For $k(1 + c^2)\|K\| < 1$, guaranteed convergence with rate $O(n^{-1/2})$.

Numerical Example: Bruck's Iterative Scheme

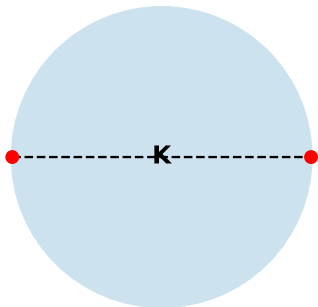
Python Implementation

```
def brucks_iteration(K_operator, N_operator, w, x0, max_iter=500, tol=1e-6):  
    """  
    Bruck Jr. iterative scheme for  $u + K*N(u) = w$   
    Returns iterates and convergence history  
    """  
    residuals = []  
    u_vals = [x0]  
  
    for n in range(1, max_iter):  
        # Apply nonlinear operator  
        N_u = N_operator(u_vals[-1])  
  
        # Apply linear operator:  $K*N(u)$   
        K_N_u = K_operator(N_u)  
  
        # Compute residual and update  
        residual = np.linalg.norm(u_vals[-1] + K_N_u - w)  
        residuals.append(residual)
```

Banach Space Properties

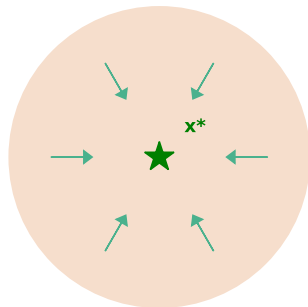
Bounded Convex Set K in Banach Space

--- Diameter $\delta(K)$



Normal Structure & Nonexpansive Mappings

★ Fixed point x^*



Kirk's Theorem and Normal Structure

Theorem 13 (Kirk's Fixed Point Theorem)

Let K be a bounded convex set in a Banach space X with normal structure, and let $F : K \rightarrow K$ be a nonexpansive mapping (i.e., $\|Fx - Fy\| \leq \|x - y\|$ for all $x, y \in K$). Then F has a fixed point in K .

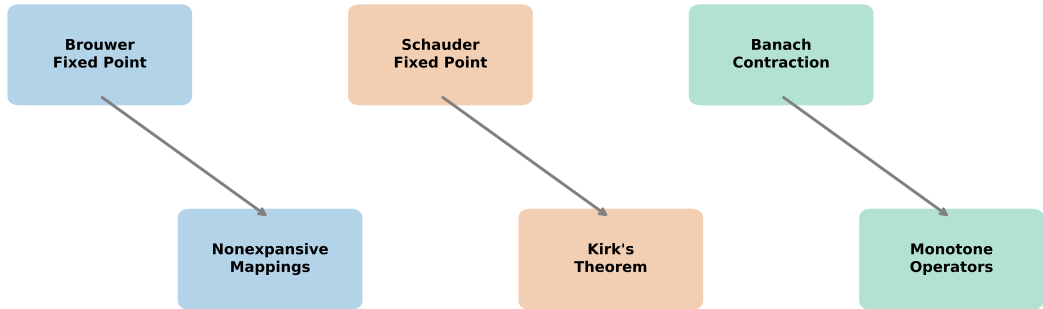
Key observations:

- Normal structure is more general than compactness
- Includes unbounded but non-trivial convex sets
- Equivalence with reflexivity + weak lower semicontinuity properties
- Uniformly convex spaces automatically have normal structure

Corollary 14

If K has normal structure and $F : K \rightarrow K$ is nonexpansive, then $\delta(K_0) < \delta(K)$ for any closed convex subset $K_0 \neq K$ that is invariant under F .

Hierarchy of Fixed Point Theorems



Required Conditions for Existence:

Uniformly Convex Spaces

Definition 15 (Uniformly Convex)

A Banach space X is **uniformly convex** if for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\left\| \frac{x+y}{2} \right\| > 1 - \delta \quad \text{whenever } \|x\| = \|y\| = 1, \|x - y\| \geq \epsilon$$

Equivalent characterizations:

- The norm is strictly convex
- Every bounded sequence has at most one weak limit point
- If $x_n \rightharpoonup x$ and $\|x_n\| \rightarrow \|x\|$, then $x_n \rightarrow x$ (norm convergence)

Theorem 16

Every uniformly convex Banach space is reflexive, and thus has normal structure.

Examples: L^p spaces with $1 < p < \infty$, Sobolev spaces $W^{1,p}$ with $1 < p < \infty$.

Main Results Summary

Problem Class	Operator Structure	Existence Tool
Hammerstein Eq.	$x + KN_f x = y$	Monotone K + Hemicontinuous N_f
Generalized Ham.	$u + K(u)N_f u = v$	Angle-bounded $K(u)$
Mixed Type	$u + \sum_j K_j N_j u = 0$	Product space structure
Urysohn Eq.	$x + \int k(s, t, x(t))dt = y$	Kernel monotonicity
Fixed Points	$F(x) = x$ in K	Normal structure

Unifying principle: Monotone operators and normal structure provide geometric and analytical tools for existence and uniqueness results across different problem classes.

Application Domains

1. Nonlinear Differential Equations:

- Boundary value problems as integral equations
- Coupled systems of PDEs

2. Nonlinear Integral Equations:

- Hammerstein equations from integral operators
- Volterra and Fredholm type problems

3. Nonlinear Optimization:

- Variational inequalities
- Monotone inclusion problems: $0 \in A(x)$

4. Applied Mathematics:

- Heat radiation transfer (Chandrasekhar's equation)
- Fluid mechanics (Navier-Stokes type problems)
- Control theory (optimal control formulations)

5. Computational Methods:

- Fixed point iterations with guaranteed convergence
- Iterative schemes with convergence rates

Open Problems and Extensions

- **Weak convergence:** Characterize spaces where weak convergence sequences have fixed points
- **Non-compact operators:** Develop theory for condensing maps and measures of noncompactness
- **Nonlinear semigroups:** Evolution equations with monotone operators
- **Variational methods:** Mountain pass theorems and critical points
- **Computational aspects:** Optimal convergence rates and algorithmic improvements

These extensions connect to variational principles, degree theory, and topological methods discussed in subsequent chapters.

References and Further Reading I

Primary Source:

- Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer. Chapter 6-8.

Classical References:

- Dolph, C. L., & Minty, G. J. (1964). On nonlinear integral equations of the Hammerstein type. *Nonlinear Integral Equations*.
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Modern Approaches:

- Brezis, H. (2011). *Functional Analysis, Sobolev Spaces and Partial Differential Equations*.
- Zeidler, E. (1990-1995). *Nonlinear Functional Analysis and its Applications* (Vol. I-V).
- Bauschke, H. H., & Combettes, P. L. (2017). *Convex Analysis and Monotone Operator Theory*.

References and Further Reading II

Related Topics:

- Leray-Schauder degree theory for existence proofs
- Condensing maps and measures of noncompactness
- Asymptotic behavior of iterative sequences
- Applications to PDEs and integral equations

Thank you!

Questions and Discussion